



End Term (Even) Semester Regular/Back Examination May 2026

Roll no. 2318517.....

Name of the Program and semester: B.Tech (CSE) and IV Semester

Name of the Course: **Microprocessors**

Course Code: **TCS-403**

Time: 3 hours

Maximum Marks: 100

Note:

- (i) All the questions are compulsory.
- (ii) Answer any two sub questions from a, b and c in each main question.
- (iii) Total marks for each question is 20 (twenty).
- (iv) Each sub-question carries 10 marks.

Q1.

(2X10=20 Marks)

- a. Describe memory segmentation in 8086 and explain physical address calculation with examples. (CO1)
- b. Explain the features of 8085 and also the interrupts of 8085 microprocessor. (CO1)
- c. Explain following pins of 8085 (CO1)
 - i) ALE ii) SID iii) TRAP iv) READY

Q2.

(2X10=20 Marks)

(CO2)

- a. Explain the meaning of following 8086 instructions
 - i) XOR AL, BH
 - ii) CMPSB
 - iii) MOV AX, [2020]
 - iv) MUL BX
 - v) CBW
- b. Explain CALL and RET instruction in 8085 with a proper example. (CO2)
- c. Write an assembly language program using 8086 for Binary to Gray conversion and explain. (CO3)

Q3.

(2X10=20 Marks)

- a. Write an assembly language program using 8085 to find the largest number in an array where the starting address of the array is 3000H. (CO3)
- b. Describe various addressing modes of 8086 with examples. (CO2)
- c. Write an assembly language program using 8085/8086 to count the number of even numbers in an array. (CO3)

Q4.

(2X10=20 Marks)

(CO4)

- a. Draw the block diagram of 8259 and explain its working. (CO4)
- b. Explain the control word register of 8255. Write an assembly language program to read the data from PA and PC_{upper} and display the output on PB and PC_{lower}. (CO4)
- c. Interface a 4KB RAM with 8085 microprocessor where the starting address of the memory is 8000H. Determine addressing range. (CO4)

Q5.

(2X10=20 Marks)

(CO5)

- a. Explain the working of ADC 0808/0809 with a program of interfacing. (CO5)
- b. Draw the block diagram of 8253/8254 timer and also briefly explain the modes. (CO6)
- c. Design a microprocessor-based system with block diagram.